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RESEARCH

Underwater image enhancement via deep modeling with frequency decomposition

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Abstract

Underwater images often suffer from spatially varying illumination, color cast, low contrast, and veiling effects caused by wavelength-dependent attenuation and scattering. To address these degradations, this paper proposes FDM-Net, a frequency-decomposition modeling network for underwater image enhancement. During the downsampling stage, FDM-Net explicitly decouples low-frequency global information and high-frequency local details, and incorporates physics-based underwater imaging cues into feature learning via a frequency-expert-guided bidirectional modulation strategy, thereby enhancing the robustness and interpretability of the model. During the upsampling stage, a multi-scale fusion scheme is adopted to reconstruct fine structures while preserving cross-scale consistency. On the T-90 and LSUI-TE datasets, FDM-Net achieves the best PSNR/SSIM scores of 23.58/0.915 and 28.18/0.925, respectively, and demonstrates superior visual enhancement performance. These results indicate that the proposed method has strong robustness and promising potential for real-world underwater vision applications.

Keywords Underwater image enhancement · Frequency decomposition · Underwater imaging model · Bidirectional modulation · Multi-scale fusion

1 Introduction

Underwater vision plays an essential role in ocean exploration, ecological monitoring, and autonomous navigation. However, underwater images are severely degraded by wavelength-dependent absorption and scattering, which jointly lead to color distortion, contrast attenuation, and detail blurring. A commonly used underwater imaging model can be formulated as

$$I_c(\mathbf{x}) = J_c(\mathbf{x})\, t_c(\mathbf{x}) + B_c(\mathbf{x})\big(1 - t_c(\mathbf{x})\big), \quad c \in \{r, g, b\}, \tag{1}$$

where $I_c(\mathbf{x})$ and $J_c(\mathbf{x})$ denote the observed and latent clean images, respectively, $B_c(\mathbf{x})$ is the background light, and $t_c(\mathbf{x})$ is the transmission map. In underwater scenes, the transmission is wavelength-dependent and can be written as

$$t_c(\mathbf{x}) = \exp\big(-\beta_c d(\mathbf{x})\big), \tag{2}$$

where $d(\mathbf{x})$ denotes scene depth and β_c is the attenuation coefficient of channel c, usually satisfying $\beta_r > \beta_g > \beta_b$. Accordingly, the latent clean image can be recovered by

$$J_c(\mathbf{x}) = \frac{I_c(\mathbf{x}) - B_c(\mathbf{x})\big(1 - t_c(\mathbf{x})\big)}{\max\big(t_c(\mathbf{x}), \epsilon\big)}, \tag{3}$$

where ϵ is a small constant for numerical stability. This formulation explains the severe red-channel attenuation, green-blue color cast, and low contrast commonly observed in underwater images. The analysis in Fig. 1 further confirms the pronounced channel imbalance and wavelength-dependent color degradation on the UIEB dataset (Li et al. 2019a). Such degradations substantially reduce the reliability of downstream tasks such as object detection, semantic segmentation, and visual navigation (Xu et al. 2023; He et al. 2024; Köser and Frese 2019).

Early underwater image enhancement (UIE) methods mainly relied on physical priors and imaging models to

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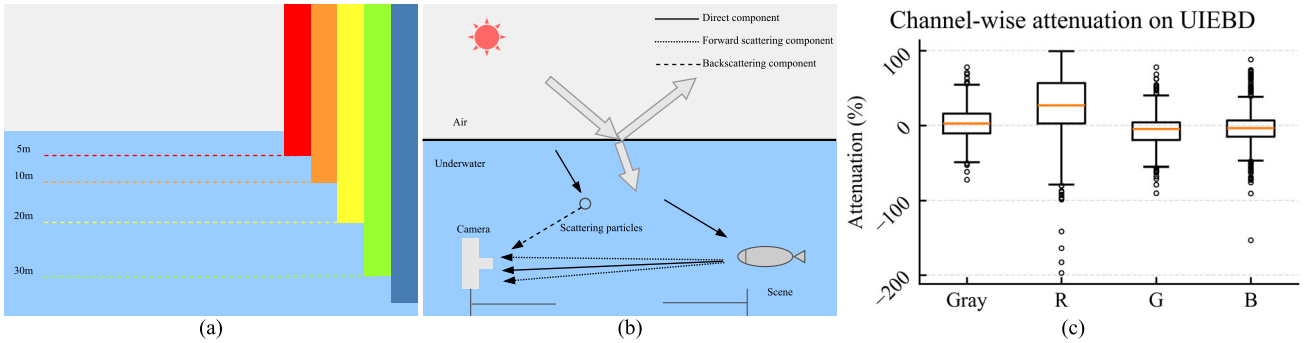


Fig. 1 Qualitative and quantitative analysis of underwater degradation. (**a**) Wavelength-dependent attenuation in seawater causes faster decay of long-wavelength red light, leading to severe color distortion. (**b**) Scattering and suspended particles induce multiple light deflections, resulting in blur and contrast loss. (**c**) Channel-wise statistics on the UIEB dataset show that the red channel exhibits a median attenuation of about 30%, while the green and blue channels present a slight negative attenuation, indicating a pronounced green-blue bias relative to the ground truth, where 0 indicates no difference

estimate transmission and background light for image restoration (Huang et al. 2018). These methods are physically interpretable, but their performance is often sensitive to prior violations and inaccurate parameter estimation under complex underwater conditions. In recent years, deep learning-based methods have achieved notable progress by learning direct mappings from degraded images to enhanced results (Wang et al. 2021; Qi et al. 2023; Jiang et al. 2023a; Mu et al. 2023; Guo et al. 2025; Fu et al. 2022a; Xing et al. 2024; Liu et al. 2020; Zhao et al. 2023; Naik et al. 2021; Fabbri et al. 2018; Li et al. 2019b). Nevertheless, most of these methods mainly operate in the spatial domain and do not explicitly model the complementary roles of low-frequency degradation cues and high-frequency structural details.

To address this issue, several frequency-based UIE methods have recently been proposed. WPFNet combines wavelet-domain detail modeling with pixel-domain compensation for color and illumination recovery (Liu et al. 2024), whereas WWPF and WFAC mainly enhance underwater images through color correction, contrast optimization, and subsequent wavelet-based multi-scale fusion (Zhang et al. 2023, 2025). These studies confirm the effectiveness of frequency decomposition for improving underwater image quality. However, most existing frequency-based UIE methods still rely on predefined wavelet decomposition and fusion strategies, limiting their adaptability to spatially non-uniform underwater degradations. In addition, their frequency restoration is only weakly coupled with underwater imaging physics, which restricts physical interpretability and robustness in challenging real-world scenes.

A related idea can be found in BMFH-Net (Sun et al. 2025), which introduces bidirectional modulation for remote sensing image dehazing. However, it is designed for atmospheric haze removal under the atmospheric scattering model (ASM), where modulation serves CNN–Transformer feature alignment. By contrast, UIE must address wavelength-dependent attenuation, cross-channel imbalance, and spatially varying scattering. Therefore, BMFH-Net cannot be directly applied to UIE.

To this end, we develop FDM-Net for UIE. We adopt a Improved Dynamic Frequency Separator (IDFS) (Cui et al. 2023) as an input-adaptive front-end to decompose degraded features into low- and high-frequency components. On this basis, we propose a Frequency-Guided Differential Expert Network , in which a gate network dynamically selects between the customized DERB (Deformable Enhanced Residual Block) and MDAB (Multi-scale Dilated Attention Block) experts to differentially model global degradation cues and local structural details in the two frequency branches. For the outputs of both branches, we further introduce a wavelength-aware UIM-based inversion to estimate physically meaningful variables such as transmission and background light, thereby improving the physical consistency of feature restoration. The resulting branch features are then fed into a bidirectional affine modulation module to promote cross-frequency interaction and collaborative restoration, while avoiding imbalanced enhancement between the two frequency components. During the upsampling stage, a Multi-scale Convolutional Fusion Network (MCFN) is employed to aggregate and refine multi-scale features for texture recovery and detail preservation. In addition, a hybrid loss is designed to jointly enforce spatial consistency and spectral fidelity. To validate the proposed method comprehensively, we conduct experiments on two full-reference UIEB (Li et al. 2019a) and LSUI datasets (Peng et al. 2023) and two no-reference U45 (Li et al. 2019c) and Challenge-60 (Li et al. 2019a), using Peak Signal-to-Noise Ratio (PSNR) (Wang et al. 2004), Structural Similarity Index Measure (SSIM) (Wang et al. 2004), Learned Perceptual Image Patch Similarity (LPIPS) (Zhang et al. 2018), Underwater Color Image Quality Evaluation (UCIQE) (Yang and Sowmya 2015), and Underwater Image Quality Measure (UIQM)

(Panetta et al. 2015) for quantitative evaluation, along with visual assessment on real-world underwater images.

The key contributions of this paper are as follows:

- We propose a Frequency-Guided Differential Expert Network, where high- and low-frequency components are modeled by our custom modules, DERB and MDAB, for more accurate underwater image restoration.
- We replace the traditional atmospheric scattering-based inversion with an UIM-based inversion, which introduces wavelength-related compensation to account for underwater light propagation effects, ensuring a more physically consistent image restoration process.
- During the upsampling stage, we introduce the MCFN to combine and refine features from different scales, enabling better texture recovery and fine detail preservation. This contributes to a more accurate and visually pleasing underwater image restoration.
- We design a hybrid loss function that optimizes both spatial consistency and spectral fidelity. This regularization ensures enhanced image quality across both spatial and frequency domains, improving color restoration and structural detail preservation in underwater images.

2 Related work

UIE aims to alleviate color distortion, low contrast, and haze caused by light absorption and scattering. Existing methods can be broadly grouped into physics-based, deep learning-based, and frequency-based approaches.

2.1 Physics-based methods

Early UIE methods mainly relied on physical models of underwater light propagation to reverse scattering and absorption. These methods usually estimate transmission maps, attenuation coefficients, or background light to restore degraded images. A representative example is the DCP proposed by He et al. He et al. (2010), which assumes that at least one color channel in a haze-free image contains low-intensity values. Peng and Cosman (2017) further introduced a restoration method based on image blurriness and light absorption, using depth estimation to better model attenuation. Histogram-based methods, such as HE (Kim 1997) and CLAHE (Pisano et al. 1998), have also been widely used to improve contrast and visibility, while Dorini and Leite (2009) extended multiscale processing to enhance underwater details. Although these methods are interpretable and computationally efficient, they heavily depend on fixed priors and assumptions, which often fail in complex underwater scenes with non-uniform degradation and diverse optical conditions (Huang et al. 2018; Iqbal et al. 2010).

2.2 Deep learning-based methods

Deep learning has significantly advanced UIE by learning complex enhancement mappings from large-scale data. Li et al. (2019b) introduced WaterNet, which highlights the importance of real-world applicability and benchmark datasets. Li et al. (2020) then proposed UWCNN, an underwater scene prior-inspired network for image enhancement. Wang et al. (2021) developed UIEC2Net to combine two color spaces for color correction, while Qi et al. (2023) incorporated Retinex theory into a multiscale enhancement network. Zhao et al. (2023) further emphasized sufficient spatial-frequency interaction for complex underwater enhancement. Unsupervised methods such as USUIR (Fu et al. 2022b) improve generalization by leveraging unpaired data. Fu et al. (2022a) introduced uncertainty-aware enhancement, and Jiang et al. (2023a) proposed Five A+, a lightweight model with only 9K parameters. Hybrid methods have also emerged: Mu et al. (2023) embedded physical priors into a dynamic enhancement framework, while Guo et al. (2025) proposed a prior-guided hybrid method. Despite their strong performance, most deep models still struggle with limited interpretability and robustness under highly varying underwater conditions.

2.3 Frequency-based methods

Recent studies have increasingly explored frequency-domain modeling, since frequency decomposition helps disentangle global color/illumination degradation from local detail loss. In UIE, Liu et al. (2024) proposed WPFNet, which combines multi-level discrete wavelet transform with progressive wavelet–pixel fusion to jointly recover details, color, and illumination. Wang et al. (2024) introduced MWEN, where low-frequency components are used for color correction and high-frequency components for sparse detail restoration. Li et al. (2025) further proposed MWPRFN, which enhances cross-frequency interaction through a multi-level wavelet pyramid and atmospheric-light compensation. Zhang et al. (2023) proposed WWPF to enhance underwater images through attenuation-guided color correction, contrast optimization, and weighted wavelet fusion, while Zhang et al. (2025) further introduced WFAC to fuse globally and locally enhanced results through wavelet decomposition. Beyond UIE, Wang et al. (2023) and Zhang et al. (2024) exploited Fourier amplitude/phase information for low-light enhancement, while Jiang et al. (2023b) and Huang et al. (2024) combined wavelet decomposition with diffusion-based restoration. In addition, Sun et al. (2025) proposed BMFH-Net for remote-sensing dehazing via heterogeneous high/low-frequency modeling. However, most existing methods still rely on predefined wavelet/Fourier decomposition and hand-crafted fusion strategies, limiting their adaptabil-

ity to spatially non-uniform underwater degradation and their coupling with underwater imaging physics.

3 Proposed methodology

As illustrated in Fig. 2, FDM-Net is proposed to address spatially non-uniform degradation and wavelength-dependent absorption in underwater images. Considering that underwater degradation is mainly caused by absorption and scattering, we first decouple degraded features into low- and high-frequency components via a dynamic frequency separator for targeted restoration. Then, frequency-expert guidance, physics-based inversion, and bidirectional affine modulation are introduced to achieve accurate and physically consistent enhancement. During upsampling, the proposed MCFN further improves texture recovery by aggregating multi-scale features while preserving fine details and structural integrity. The details of each module are described below.

3.1 Improved dynamic frequency separator

The degradation process in underwater images is primarily influenced by light absorption and scattering, with low-frequency components being more affected by light attenuation, while high-frequency components are more influenced by particle scattering. To separate the low- and high-frequency components of degraded images for targeted restoration, we adopt the DFS module (Cui et al. 2023) with a minor modification by introducing Global Max Pooling (GMP). The term "dynamic" is inherited from the original DFS module (Cui et al. 2023), where the frequency decomposition is input-adaptive rather than fixed. In our implementation, we retain this adaptive decomposition strategy and further enhance it by introducing GMP to complement GAP for richer global cues. Based on this design, the input feature $X_{in} \in \mathbb{R}^{H\times W\times C}$ is adaptively decomposed into low- and high-frequency components through a soft allocation strategy:

$$X_l, X_h = \mathrm{IDFS}(X_{in}), \tag{4}$$

where X_l and X_h denote the low- and high-frequency features, respectively. To further refine the decoupled representations, we employ a CNN to encode the low-frequency feature X_l into L_{in}, and a Multi-head Self-Attention module to transform the high-frequency feature X_h into H_{in}. This low–high frequency decoupling design provides a structured feature foundation for the subsequent frequency differential modulation process.

3.2 Frequency-expert-guided bidirectional modulation

As illustrated in Fig. 2(b), the Frequency-Expert-Guided Bidirectional Modulation (FEGBM) is composed of three key components: the Frequency-Guided Differential Expert Network (FGDE-Net), Physics-based UIM Inversion Mechanism (PUIM), and Bidirectional Affine Modulation Module (BAMM). In the following, we introduce each of these modules in detail.

3.2.1 Frequency-guided differential expert network

As shown in Fig. 3(a), underwater image degradation presents distinct characteristics in the low- and high-frequency

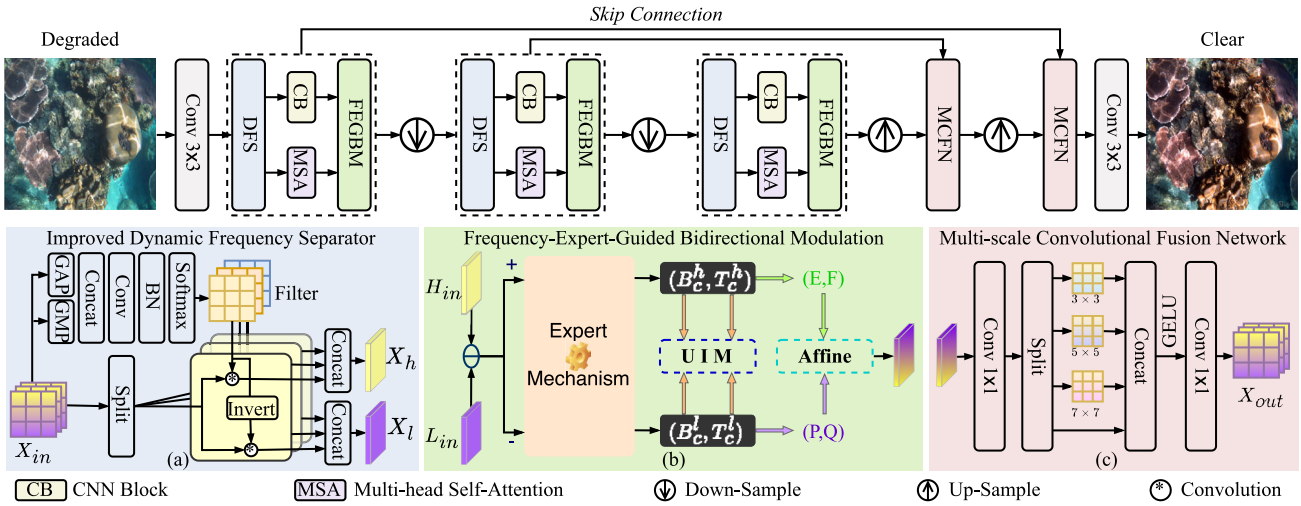


Fig. 2 Overview of the proposed FDM-Net. The core components of FDM-Net include the Improved Dynamic Frequency Separator (IDFS), the Frequency-Guided Bidirectional Modulation (FEGBM), and the Multi-scale Convolutional Fusion Network (MCFN). In particular, FEGBM consists of a frequency differential expert module, a physics-guided underwater imaging module, and a bidirectional affine modulation module

domains. To model such frequency-specific degradation, we design a FGDE-Net with two parallel branches, i.e., a high-frequency differential expert and a low-frequency differential expert. Each branch is composed of multiple expert blocks, where DERBs are used for the high-frequency branch to enhance texture and edge details, while MDABs are adopted for the low-frequency branch to capture broader contextual and structural information.

To further characterize the residual degradation between the decoupled features, we construct bidirectional differential features from the input high-frequency component H_{in} and low-frequency component L_{in} as

$$H_d = H_{in} - L_{in}, \quad L_d = |L_{in} - H_{in}|, \tag{5}$$

where H_d emphasizes texture and edge residuals, and L_d captures illumination variations and structural discrepancies. By explicitly modeling the difference between low- and high-frequency representations, this design reduces redundancy and promotes selective cross-frequency interaction, thereby facilitating the modeling of distinct underwater degradation effects, such as wavelength-dependent absorption in low-frequency components and scattering-induced corruption in high-frequency details.

High-Frequency Differential Expert As shown in Fig. 3(a), the high-frequency differential expert branch is designed to recover scattering-corrupted textures and edges through differential feature modeling and adaptive expert selection. By employing DERBs, this branch enhances high-frequency details that are severely degraded by forward and backward light scattering. Figure 4 illustrates the forward computation of the i-th DERB ($i = 1, 2, \ldots, N$) given the high-frequency differential feature $H_d \in \mathbb{R}^{H\times W\times C}$. Each DERB first applies a lightweight 3×3 convolution to predict spatial offsets:

$$\Delta p_i = \mathrm{Conv}_{3\times3}(H_d), \quad \Delta p_i \in \mathbb{R}^{H\times W\times 2P}, \tag{6}$$

where $P = k^2$ denotes the number of sampling locations in a $k \times k$ convolution kernel. The predicted offsets are then used for deformable convolution, followed by a 1×1 convolution and residual addition:

$$F_i = \mathrm{Conv}_{1\times1}\big(\mathrm{DeformConv}(H_d, \Delta p_i)\big) + H_d. \tag{7}$$

To improve efficiency and stability, a gating network is introduced to adaptively select the most relevant DERBs according to global descriptors extracted from H_d:

$$H_e = \mathrm{GAP}(H_d) + \mathrm{GMP}(H_d), \quad H_e \in \mathbb{R}^{1\times1\times C}, \tag{8}$$

$$W_h = \sigma\Big(T_k\big(\tau_1(H_e M_p) + \tau_2(\mathrm{SoftPlus}(H_e M_q))\big)\Big), \quad W_h \in \mathbb{R}^N, \tag{9}$$

where M_p and M_q are learnable weight matrices, and τ_1 and τ_2 denote fully connected layers. The final enhanced high-

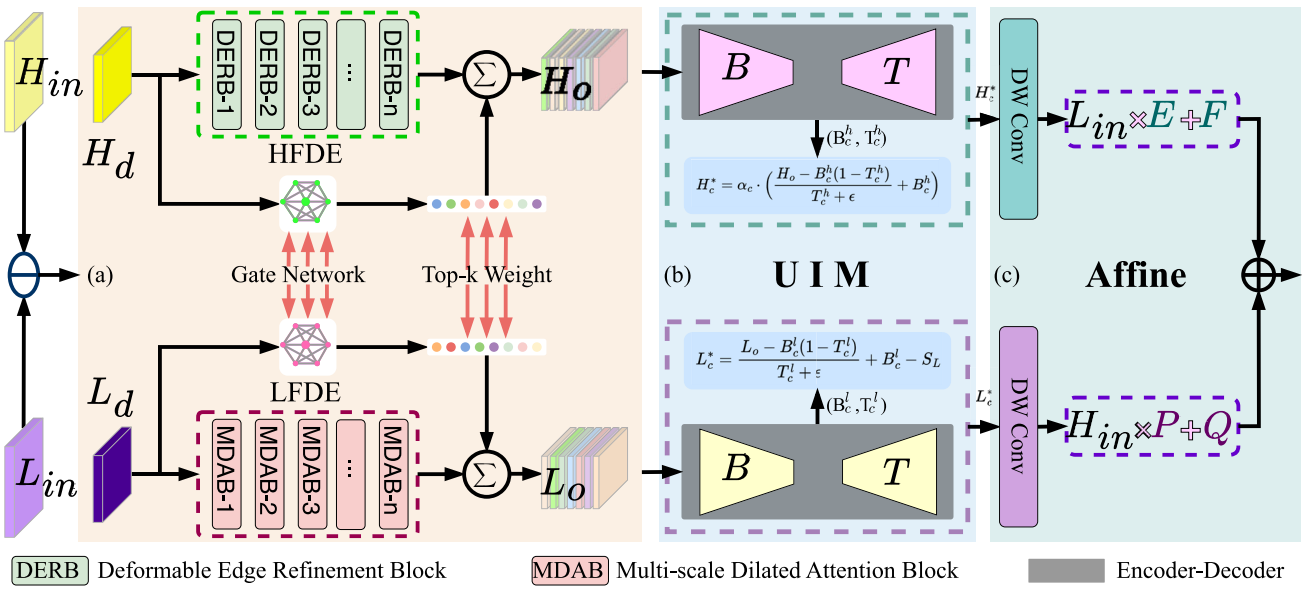


Fig. 3 (**a**) Frequency-Guided Differential Expert Network (FGDE-Net). (**b**) Physics-based UIM Inversion Mechanism (PUIM), where an encoder-decoder network is employed to learn the inversion parameters B and T in the UIM model. (**c**) Bidirectional Affine Modulation Module (BAMM), where DWConv layer generate the high-frequency branch coefficients E and F, and the low-frequency branch coefficients P and Q

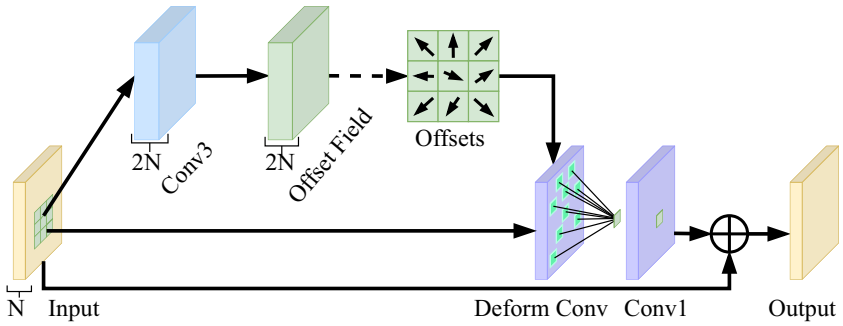


Fig. 4 Overview of the proposed Deformable Enhancement Residual Block. Conv3 denotes a 3×3 convolution layer that expands the input features (from N to $2N$ channels). Offsets represent the learnable spatial sampling offsets for the Deform Conv, which are predicted by the offset-field branch.Conv1 denotes a 1×1 convolution layer that serves as a standard convolution to refine the features after the Deform Conv

frequency feature is obtained by

$$H_o = \sum_{i=1}^{N} W_h^i \cdot F_i, \quad H_o \in \mathbb{R}^{H \times W \times C}. \tag{10}$$

This design enables adaptive high-frequency restoration by activating the most relevant experts, and provides detail-enhanced features for subsequent reconstruction and supervision.

Low-Frequency Differential Expert Due to the significant impact of light absorption and scattering on low-frequency components, severe degradation occurs in color balance, contrast, and overall structure. Therefore, as shown in Fig. 3(a), the low-frequency differential expert branch employs MDABs to capture global structure and illumination information, thereby compensating for wavelength-dependent attenuation and facilitating balanced restoration. Figure 5 shows the forward computation for the j-th MDAB ($j = 1, 2, \ldots, N$) given the low-frequency feature $L_d \in \mathbb{R}^{H \times W \times C}$. Each MDAB first uses depthwise dilated convolutions to extract multi-scale contextual priors and enlarge the receptive field with low computational cost:

$$F_{1,j} = \delta\big(\mathrm{DWConv}_d(L_d)\big), \tag{11}$$

$$F_{2,j} = \delta\big(\mathrm{DWConv}_d(F_{1,j})\big), \tag{12}$$

where DWConv_d denotes depthwise convolution with dilation rate d (see Fig. 6), and $\delta(\cdot)$ is a nonlinear activation function. A channel-attention module is then applied to refine the feature response, yielding

$$F_j = \mathrm{CA}(F_{2,j}) + L_d. \tag{13}$$

Following the high-frequency branch, the low-frequency branch adopts the same gating strategy to generate expert weights W_l, so that the most informative MDABs are selectively activated. The final low-frequency feature is obtained as

$$L_o = \sum_{j=1}^{N} W_l^j \cdot F_j, \quad L_o \in \mathbb{R}^{H \times W \times C}. \tag{14}$$

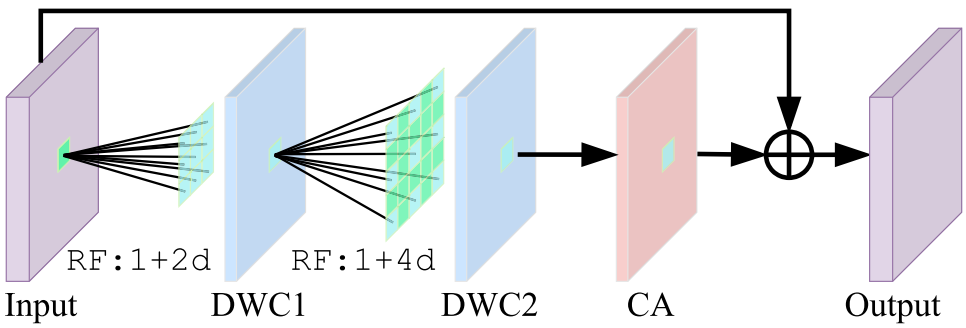


Fig. 5 Overview of the proposed Multi-scale Dilated Attention Block. DWC1 and DWC2 denote two cascaded depthwise dilated convolution layers that progressively enlarge the receptive field (RF: $1 + 2d$ and $1 + 4d$) to capture multi-scale contextual priors. CA denotes a channel-attention module that aggregates global radiance statistics and adaptively reweights the feature channels before residual fusion with the input

3.2.2 Physics-based UIM inversion mechanism

Underwater image degradation is mainly caused by wavelength-dependent absorption and scattering, which result in color distortion, contrast attenuation, and detail loss. To explicitly incorporate this physical prior into restoration, we introduce a PUIM module. Instead of directly learning the restored features in a purely data-driven manner, this module performs feature correction by following the underwater imaging model and its inverse formulation in (1) and (3), as illustrated in Fig. 3(b). Specifically, an encoder-decoder backbone is employed to estimate the channel-wise background light and pixel-wise transmission for the high- and low-frequency branches, respectively:

$$B_c^h = \mathrm{Softmax}(\mathrm{MLP}(\mathrm{GAP}(H_o))), \tag{15}$$

$$T_c^h = \sigma\big(\mathrm{Conv}_{1\times1}(\mathrm{EncDec}(\mathrm{Conv}_{1\times1}(H_o)))\big), \tag{16}$$

$$B_c^l = \mathrm{Softmax}(\mathrm{MLP}(\mathrm{GAP}(L_o))), \tag{17}$$

$$T_c^l = \sigma\big(\mathrm{Conv}_{1\times1}(\mathrm{EncDec}(\mathrm{Conv}_{1\times1}(L_o)))\big). \tag{18}$$

To account for wavelength-dependent attenuation, we further introduce a learnable channel-wise compensation vector $\alpha_c \in \mathbb{R}^3$, initialized as $\alpha_c = [1.5,\ 1.0,\ 0.8]$, where the red channel is amplified to compensate for severe absorption, while the blue-green dominance is moderately suppressed. Based on the estimated physical parameters, the high- and low-frequency features are restored through branch-specific inversion:

$$H_c^* = \alpha_c \cdot \Big(\frac{H_o - B_c^h(1 - T_c^h)}{T_c^h + \epsilon} + B_c^h\Big), \tag{19}$$

$$L_c^* = \frac{L_o - B_c^l(1 - T_c^l)}{T_c^l + \epsilon} + B_c^l - S_L, \tag{20}$$

where S_L denotes a low-frequency scattering correction term computed from the spatially averaged low-frequency response. The compensation vector α_c is only applied to the high-frequency branch, since wavelength-dependent absorption mainly affects reflected detail components, while the low-frequency branch primarily encodes global illumination and scattering. This design enables physically consistent restoration while preserving the distinct degradation characteristics of the two frequency streams.

3.2.3 Bidirectional affine modulation module

As shown in Fig. 3(c), although the physics-guided inversion yields physically corrected high- and low-frequency features, directly fusing them may still cause imbalance between global illumination and fine details. To address this issue, we design a BAMM, which generates branch-specific affine parameters and performs residual cross-modulation between the two frequency streams. Specifically, the inverted high- and low-frequency features H_c^* and L_c^* are fed into two depthwise separable convolution branches to predict the affine parameters:

$$[E, F] = \phi_h(H_c^*), \qquad [P, Q] = \phi_l(L_c^*), \tag{21}$$

where $\phi_h(\cdot)$ and $\phi_l(\cdot)$ denote lightweight depthwise separable convolution mappings. Here, (E, F) are used to recalibrate the low-frequency stream, while (P, Q) are used to refine the high-frequency stream. The bidirectional affine modulation is given by

$$L_r = L_{in} \odot E + F, \tag{22}$$

$$H_r = H_{in} \odot P + Q, \tag{23}$$

where $\odot$ denotes element-wise multiplication. In this way, low-frequency structural cues guide high-frequency refinement, while high-frequency responses provide complementary detail-aware calibration for the low-frequency branch. The final output is obtained by residual fusion:

$$O = L_r + H_r. \tag{24}$$

Through this design, the two frequency streams are coupled cooperatively, enabling the network to jointly preserve global structural consistency and fine-grained texture fidelity in challenging underwater scenes.

3.3 Multi-scale convolutional fusion network

During the upsampling stage, recovering high-resolution structures and fine textures is crucial for perceptually faithful underwater image reconstruction. Although the frequency-expert guided modulation provides physically consistent restoration, it is less effective at refining local spatial details during progressive upsampling. To address this issue, we introduce a MCFN to enhance spatial fidelity and improve edge and texture recovery. As shown in Fig. 2(c), MCFN employs three parallel depthwise separable convolution branches with different kernel sizes to capture multi-scale spatial cues (Szegedy et al. 2016). Given the upsampled feature map X_o, which integrates the outputs of the second and third FEGBM modules via a skip connection, we first project it into a multi-scale latent space:

$$X_{ms} = \mathrm{Conv}_{1\times1}(X_o). \tag{25}$$

The projected feature is evenly split into three channel groups, which are processed by depthwise convolutions with progressively larger kernels:

$$X_i = \mathrm{DWConv}_{k_i\times k_i}\big(X_{ms}^{(i)}\big), \quad i \in \{1, 2, 3\}, \tag{26}$$

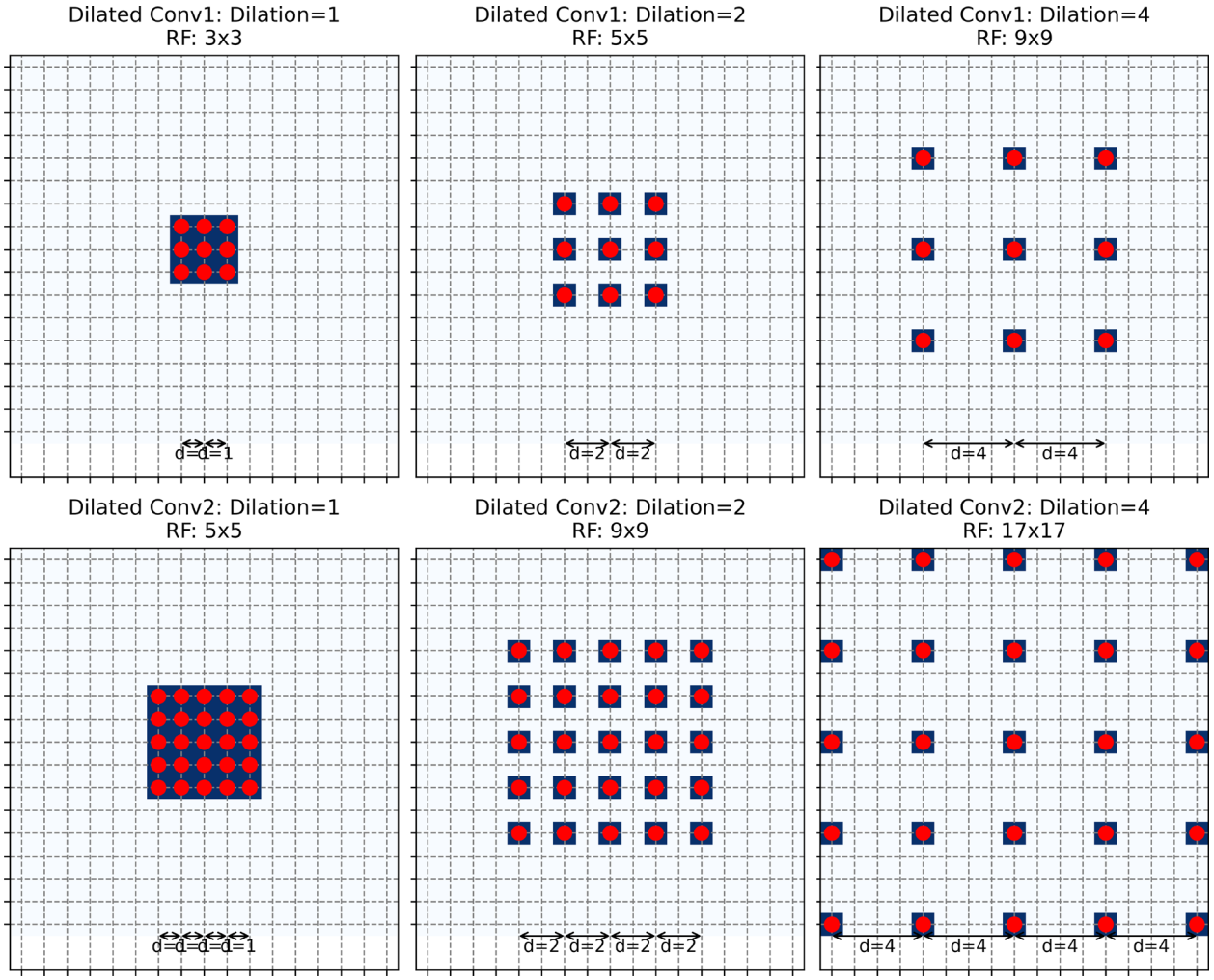


Fig. 6 Visualization of receptive field expansion with systematic dilated convolutions. For the first Dilated Conv1, dilation rates of 1, 2 and 4 yield receptive fields of 3×3, 5×5, and 9×9, respectively. Stacking a second Dilated Conv2 further enlarges the receptive field to 5×5, 9×9, and 17×17 under the same dilation settings. The receptive field grows exponentially while the number of parameters remains unchanged, enabling efficient multi-scale context aggregation

where $(k_1, k_2, k_3) = (3, 5, 7)$. These multi-scale features are then fused with the original feature X_o to preserve structural priors, and the refined output is obtained by

$$X_{out} = \mathrm{Conv}_{1\times1}\Big(\mathrm{GELU}\big(\mathrm{Concat}(X_1, X_2, X_3, X_o)\big)\Big). \quad (27)$$

This design enables MCFN to effectively refine contour continuity and texture fidelity while suppressing interpolation artifacts. Meanwhile, the depthwise separable formulation keeps the module lightweight, making it suitable for high-resolution underwater image restoration.

3.4 Loss function

The loss design follows the restoration pipeline of FDM-Net. Specifically, IDFS decomposes degraded features into low- and high-frequency components, while FEGBM performs targeted restoration on the two frequency streams via frequency-guided experts, physics-based inversion, and bidirectional affine modulation. During upsampling, MCFN further refines local structures and texture details. Accordingly, we adopt a hybrid loss function that optimizes spatial fidelity and frequency-spectrum consistency:

$$\mathcal{L}_{\text{total}} = \alpha\,\mathcal{L}_{\text{spatial}} + \beta\,\mathcal{L}_{\text{wave}} + \gamma\,\mathcal{L}_{\text{fourier}}, \quad (28)$$

where α, β, and γ are balancing coefficients. In our experiments, we set $\alpha = 1.0$, $\beta = 0.5$, and $\gamma = 0.1$.

We use the smooth L_1 loss as the basic discrepancy measure:

$$\mathcal{L}_s(A, B) = \frac{1}{N}\sum_{i=1}^{N}\begin{cases}0.5(A_i - B_i)^2, & \text{if } |A_i - B_i| < 1,\\ |A_i - B_i| - 0.5, & \text{otherwise,}\end{cases} \quad (29)$$

where N denotes the total number of elements in A and B. The spatial-domain loss directly constrains the restored image $\hat{X}$ against the ground truth X:

$$\mathcal{L}_{\text{spatial}}(\hat{X}, X) = \mathcal{L}_s(\hat{X}, X). \tag{30}$$

It serves as the primary supervision for overall color, luminance, and structural reconstruction.

To match the frequency-decoupled design of FDM-Net, we further introduce a wavelet-domain loss. Let

$$\text{DWT}(X) = \left(X_L, \{X_H^m\}_{m\in\{h,v,d\}}\right), \tag{31}$$

where X_L denotes the low-frequency sub-band and X_H^m denotes the horizontal, vertical, and diagonal high-frequency sub-bands. The wavelet-domain loss is defined as

$$\mathcal{L}_{\text{wave}}(\hat{X}, X) = \mathcal{L}_s(\hat{X}_L, X_L) + \sum_{m\in\{h,v,d\}} \mathcal{L}_s(\hat{X}_H^m, X_H^m), \tag{32}$$

which explicitly supervises low-frequency luminance/structure recovery and high-frequency detail reconstruction.

Finally, to regularize the global spectral distribution and suppress frequency distortions introduced during restoration and upsampling, we employ a Fourier-domain consistency term:

$$\mathcal{L}_{\text{fourier}}(\hat{X}, X) = \left\| \mathcal{F}(\hat{X}) - \mathcal{F}(X) \right\|_1, \tag{33}$$

where $\mathcal{F}(\cdot)$ denotes the 2D FFT. This term complements the bidirectional affine modulation and MCFN by encouraging sharper spectral reconstruction and reducing interpolation-induced artifacts.

Overall, the hybrid objective is tightly coupled with the architecture design: $\mathcal{L}_{\text{spatial}}$ ensures global reconstruction fidelity, $\mathcal{L}_{\text{wave}}$ supervises the low-/high-frequency restoration process, and $\mathcal{L}_{\text{fourier}}$ enhances spectral consistency and fine-detail preservation. Together, they enable FDM-Net to achieve physically plausible and detail-preserving underwater image restoration.

4 Experiments

To thoroughly demonstrate the superiority and generalization ability of FDM-Net, we conduct a comprehensive series of experiments. This section presents the implementation details, the feature-map and frequency-domain analyses of FDM-Net, comparisons with SOTA methods, application validations, and ablation studies.

4.1 Experimental details

This subsection outlines the implementation setup, the datasets used for training and testing, and the evaluation criteria adopted for quantitative analysis.

4.1.1 Datasets

Reference datasets The UIEB and LSUI datasets contain paired images that enable supervised training and direct quantitative evaluation. UIEB dataset includes 950 real underwater images, among which 890 have corresponding high-quality reference counterparts. Following our split, we construct UIEB-T as the training set with 800 paired images and T-90 as the test set with 90 paired images. Similarly, LSUI provides 4279 paired images, from which we build LSUI-T with 3852 pairs for training and LSUI-TE with 427 pairs for testing.

No-reference datasets To further evaluate the generalization capability of our model under diverse and challenging real-world underwater conditions, we additionally conduct experiments on U45 and the Challenge-60 datasets. Both datasets are unpaired and are therefore widely used for no-reference quality assessment.

4.1.2 Implementation setup

All experiments were implemented in PyTorch 2.4 with CUDA 12.1 acceleration and executed on one NVIDIA RTX 4090 GPU. The model was optimized using the AdamW optimizer, with hyperparameters fixed as $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The base learning rate was initialized to 2×10^{-4} and dynamically adjusted throughout training using the Cosine Annealing Learning Rate Scheduler (Liu 2022), which gradually decreased the rate from 1×10^{-4} to 1×10^{-6}. The model was trained for 200 epochs on the UIEB-T dataset (Li et al. 2019a) and 400 epochs on the LSUI-T dataset (Peng et al. 2023). Each input image was uniformly resized to 256×256 pixels, with a batch size of 8.

4.1.3 Evaluation metrics

All methods are evaluated using full-reference, no-reference, and efficiency metrics to provide a comprehensive comparison of enhancement quality and computational performance.

(1) Full-reference metrics We employed PSNR, SSIM, and LPIPS for pixel-, structure-, and perception-level comparisons. PSNR quantifies the signal fidelity between restored

and ground-truth images:

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right), \tag{34}$$

where MSE denotes the mean squared error between corresponding pixels. SSIM assesses the consistency of luminance, contrast, and structure between two images:

$$\mathrm{SSIM}(X,Y) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2)}. \tag{35}$$

Higher PSNR/SSIM and lower LPIPS values imply better restoration quality.

(2) No-reference metrics For datasets without ground-truth references, we adopted UCIQE and UIQM. UCIQE evaluates image colorfulness, contrast, and saturation, focusing on chromatic balance. UIQM measures sharpness, contrast, and natural color distribution.

(3) Computational efficiency We further assessed model efficiency using Multiply-Accumulate Operations (MACs) (Molchanov et al. 2016), parameters (Params), and end-to-end inference time (E2E. Time). MACs reflect computational complexity and inference cost, while Params measure the model's capacity. E2E. Time reports the overall latency from input to output under a fixed hardware and implementation setting. A model achieving high performance with low MACs, fewer parameters, and reduced E2E. Time demonstrates a favorable balance between accuracy and efficiency

4.2 Feature map changes and frequency analysis

In UIE, frequency-domain information is crucial for restoring image quality, especially for correcting color distortion and low contrast. Figure 7 shows the intermediate feature maps of FDM-Net and their corresponding amplitude spectra, illustrating the separation, modulation, and fusion of low-frequency and high-frequency information at different stages. Since the first downsampling stage sufficiently reflects the variation trend, we focus on it for analysis.

During the first downsampling, the IDFS module separates the input features into low-frequency and high-frequency components, as shown in the second column of Fig. 7. The low-frequency branch mainly preserves global brightness and color information, with energy concentrated near the spectrum center, while the high-frequency branch captures details and textures, showing a more dispersed energy distribution. This demonstrates that IDFS effectively disentangles global structure and local detail information. Within FEGBM, frequency-differential expert modeling further enhances the complementary roles of the two branches.

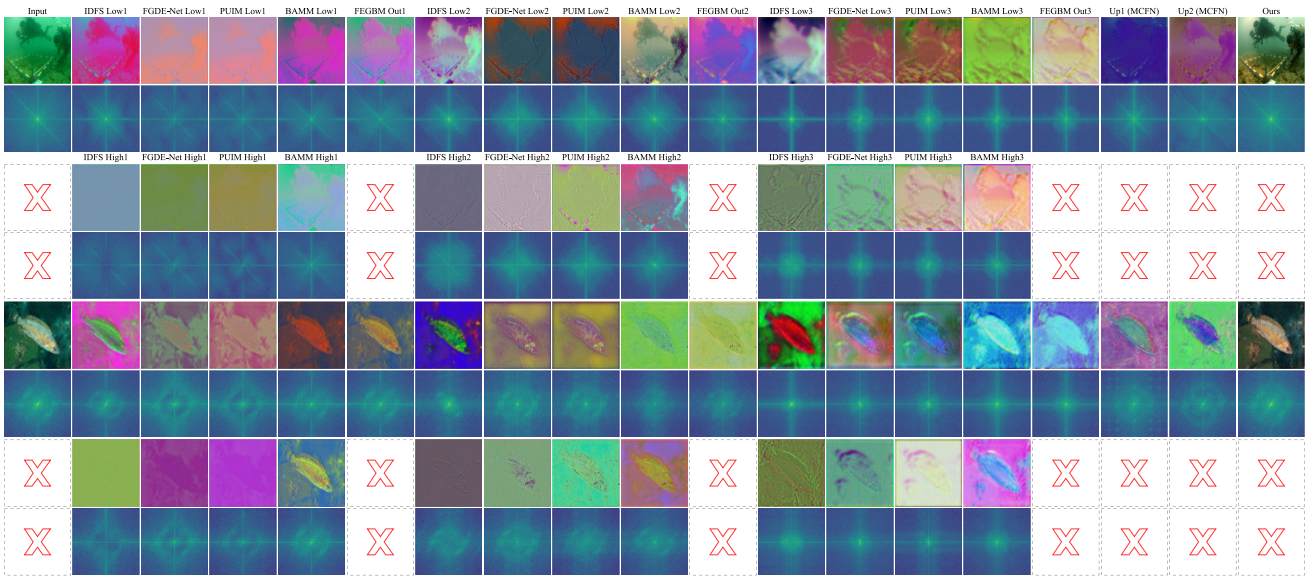


Fig. 7 Intermediate feature maps of the FDM-Net model and their corresponding amplitude spectra. The first row's example is from the T-90 dataset, and the fifth row's example is from the LSUI-TE dataset. The first column shows the original input image and its amplitude spectrum; the second to sixth columns show the feature maps and amplitude spectra during the first downsampling process, including the IDFS, FGDE-Net, PUIM, BAMM, and the output after fusion; the seventh to eleventh columns show the corresponding feature maps and amplitude spectra during the second downsampling process; the twelfth to sixteenth columns show the feature maps and amplitude spectra at different stages in the latent layer; the seventeenth and eighteenth columns show the feature maps and amplitude spectra after the first and second upsampling; the twentieth column shows the final enhanced image and its amplitude spectrum

As shown in the third to fifth columns of Fig. 7, the low-frequency features gradually incorporate more refined details, while the high-frequency features progressively recover clearer global structure. The corresponding amplitude spectra indicate increasing interaction between low-frequency and high-frequency components. In particular, the physical underwater imaging model improves local detail recovery while preserving global consistency, and the learnable bidirectional affine modulation further strengthens cross-frequency interaction and feature refinement.

After low-frequency and high-frequency fusion, the output feature map in the sixth column of Fig. 7 exhibits both clear global structure and enhanced local detail. Its amplitude spectrum becomes more balanced and comprehensive, indicating effective fusion of complementary frequency information. These observations show that FEGBM successfully integrates frequency-differential expert modeling, physical imaging guidance, and bidirectional affine modulation to improve structural and detail restoration.

During upsampling, the multi-scale MCFN progressively reconstructs high-resolution features. As shown in the seventeenth and eighteenth columns of Fig. 7, both global structure and local details become clearer and more refined. The corresponding amplitude spectra show enhanced central and surrounding energy, indicating more complete recovery of edges, textures, and fine details. Overall, the joint analysis of intermediate feature maps and amplitude spectra demonstrates that FDM-Net effectively restores image color, texture, and details through frequency decomposition, cross-frequency interaction, and progressive reconstruction, while maintaining both physical consistency and visual naturalness.

4.3 Comparison with SOTA methods

In this section, we comprehensively compare the performance of the FDM-Net method from three perspectives: qualitative analysis, quantitative evaluation, and efficiency assessment.

4.3.1 Qualitative analysis

Figure 8 first compares the color histograms of the underwater images before and after enhancement. These results indicate that the enhanced images exhibit richer details, more realistic colors, and more balanced histograms. To conduct a more comprehensive qualitative analysis, as shown in Fig. 9, we compare the proposed FDM-Net with fifteen representative SOTA UIE methods through visual examples and analyze the corresponding RGB-channel pixel distributions and proportions. Clearly, in the reference datasets T-90 and LSUI-TE, FDM-Net outperforms other methods in enhancing images affected by suspended particles, blue color casts, green color casts in deep-sea scenes, and green color casts in shallow-water scenes. For instance, as shown in the first row of Fig. 9 I, in a scene affected by suspended particles, the underwater coral plant generated by FDM-Net exhibits clearer structure and color, closely matching the Ground Truth. Similarly, as shown in the second row of Fig. 9 II, in a shallow-water scene, FDM-Net produces a clearer texture of the fish carcass. Additionally, in the corresponding pixel-frequency line charts, the images generated by FDM-Net show a trend that is more consistent with the Ground Truth's line chart. In the more challenging no-reference image datasets, U45 and Challenge-60, the FDM-Net method provides a more comprehensive and accurate real-scene reproduction, with its enhancement effects and RGB channel percentages approaching the performance levels of the current SOTA methods, GuidedHybSensUIR and FIVE A+. For example, as shown in the first row of Fig. 9 III, in a deep-sea scene, FIVE A+ generates the best visual effect of the shell, followed by FDM-Net and GuidedHybSensUIR. These results demonstrate that FDM-Net exhibits strong adaptability and effectiveness across various underwater scenes, especially in complex environments.

4.3.2 Quantitative evaluation

We evaluate FDM-Net against recent SOTA UIE methods using five metrics on two reference datasets and two no-reference datasets, as shown in Table 1.

On the two reference datasets, FDM-Net achieves the best PSNR and SSIM, demonstrating its strong capability for faithful image reconstruction. Specifically, on T-90, FDM-Net surpasses GuidedHybSensUIR by 0.22 dB in PSNR and 0.006 in SSIM. On LSUI-TE, it further improves PSNR and SSIM by 0.08 dB and 0.008, respectively. These results verify the effectiveness of the proposed frequency-decoupled restoration strategy in improving structural consistency and reconstruction quality.

Although FDM-Net does not achieve the best LPIPS, it still ranks among the top three on both reference datasets. This is mainly because our method emphasizes faithful restoration through spatial- and frequency-domain supervision, rather than directly optimizing perceptual similarity. As a result, FDM-Neft tends to preserve more stable structures and more natural color transitions. A similar trend is observed for UIQM on the reference datasets, where FDM-Net ranks second on both T-90 and LSUI-TE, indicating a good balance between visual quality and reconstruction fidelity.

On the no-reference datasets U45 and Challenge-60, we directly test FDM-Net using the weights trained on T-90 without further fine-tuning. Even under this cross-dataset setting, FDM-Net still shows competitive generalization ability. It ranks second in both UCIQE and UIQM on U45, and

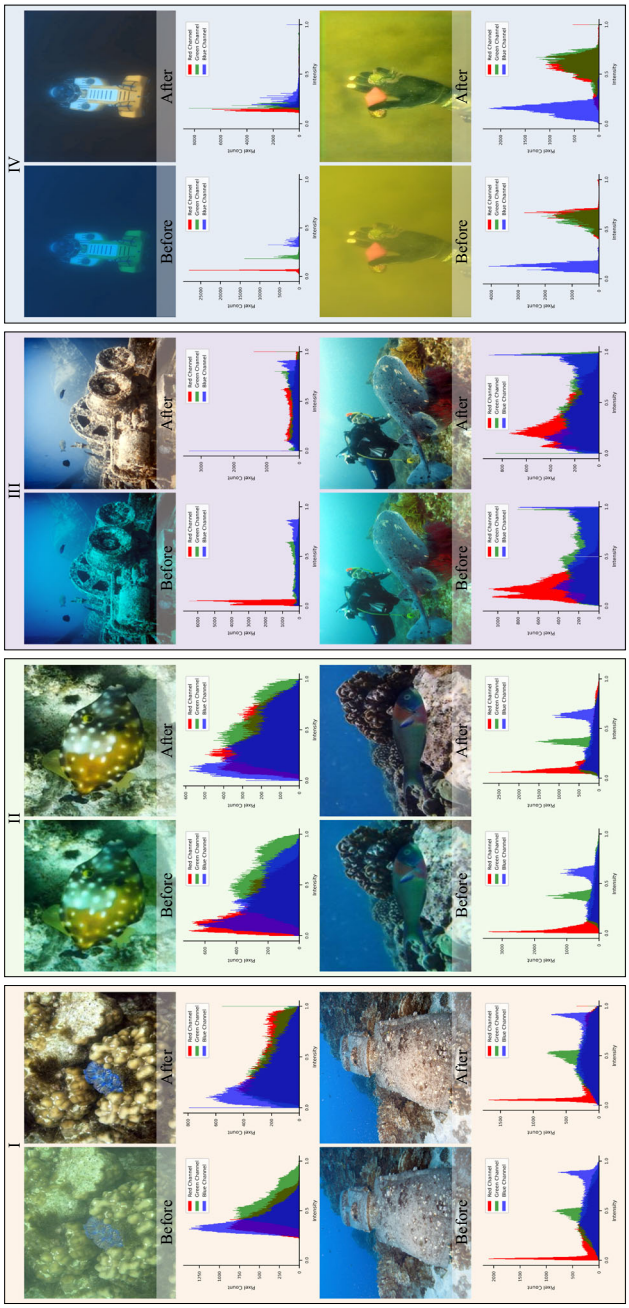


Fig. 8 Visual comparison before and after processing by FDM-Net, with the corresponding histogram shown below each image. I indicates two examples from the T-90 test set; II indicates two examples from the LSUI-TE test set; III indicates two examples from the U45 dataset; IV indicates two examples from the Challenge-60 dataset



Fig. 9 Visual comparison with SOTA methods on various benchmark datasets. (I) Results from two samples of the T-90 dataset, one affected by suspended particles and the other by a blue color cast; (II) Results from two samples of the LSUI-TE dataset, one with a green color cast and the other in a shallow-water scene; (III) Test results on the U45 dataset using the T-90 dataset pre-trained model, including one deep-sea low-brightness green color cast scene and one deep-sea low-brightness blue color cast scene; (IV) Test results on the Challenge-60 dataset using the T-90 dataset pre-trained model, showing one shallow-water green color cast scene and one deep-sea scene with insufficient lighting and severe color attenuation

Table 1 Combined quantitative results on the T-90, LSUI-TE, U45, and Challenge-60 datasets. Best results in red, second-best in blue, and third-best in purple

Methods	Year	Journal/Conf.	T-90				LSUI-TE				U45		Challenge-60	
			PSNR↑	SSIM↑	LPIPS↓	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	UIQM↑	UCIQE↑	UIQM↑	UCIQE↑	UIQM↑
WaterNet (Li et al. 2019b)	2019	TIP	21.96	0.845	0.149	3.67	20.58	0.813	0.181	3.03	0.42	3.66	0.37	3.86
UIEC2Net (Wang et al. 2021)	2021	SPIC	20.11	0.892	0.111	3.66	23.48	0.903	0.070	3.68	0.44	3.64	0.42	3.79
SCNet (Liu et al. 2020)	2022	ICASSP	22.19	0.895	0.093	3.26	25.27	0.883	0.072	3.11	0.37	2.85	0.36	3.85
GUPDM (Mu et al. 2023)	2023	ACM MM	21.69	0.853	0.130	3.19	24.81	0.868	0.122	3.19	0.25	3.67	0.25	3.80
Shallow-UWnet (Naik et al. 2021)	2021	AAAI	19.21	0.845	0.145	3.78	21.59	0.845	0.109	3.73	0.36	3.70	0.35	3.73
UWCNN (Li et al. 2020)	2020	PR	18.86	0.859	0.122	3.85	22.89	0.886	0.118	3.84	0.36	3.72	0.35	3.81
CCMSRNet (Qi et al. 2023)	2023	TGRS	22.31	0.880	0.105	3.71	27.42	0.904	0.086	3.66	0.33	3.34	0.38	3.75
USUIR (Fu et al. 2022b)	2022	AAAI	18.69	0.776	0.175	3.62	18.34	0.778	0.250	3.47	0.36	3.55	0.36	3.64
SFGNet (Zhao et al. 2023)	2023	TIP	21.49	0.896	0.134	3.03	25.13	0.874	0.127	3.10	0.45	3.53	0.41	1.07
FIVE A+ (Jiang et al. 2023a)	2023	BMVC	22.71	0.904	0.086	3.94	25.55	0.903	0.103	3.78	0.44	3.89	0.43	4.17
PUIE-Net (Fu et al. 2022a)	2022	ECCV	19.48	0.822	0.133	3.79	26.99	0.899	0.057	3.86	0.32	3.73	0.31	3.82
BMFH-Net (Sun et al. 2025)	2025	TCSVT	21.68	0.878	0.116	4.09	24.88	0.891	0.123	3.93	0.39	3.93	0.41	4.09
GuidedHybSensUIR (Guo et al. 2025)	2025	TCSVT	23.36	0.909	0.072	3.81	28.10	0.917	0.045	3.71	0.43	3.77	0.41	3.87
Ours (FDM-Net)	–	–	23.58	0.915	0.074	3.98	28.18	0.925	0.064	3.92	0.44	3.88	0.40	3.91

achieves the third-best UIQM on Challenge-60. Although its UCIQE on Challenge-60 is not the highest, this is reasonable because no-reference metrics are often sensitive to aggressive enhancement, such as stronger contrast, saturation, and sharpness. In contrast, FDM-Net adopts a more stable restoration strategy, which helps suppress over-enhancement artifacts and preserve a more natural visual appearance under domain shifts.

Overall, FDM-Net achieves balanced performance across different datasets and evaluation settings. These results suggest that FDM-Net is particularly suitable for underwater applications requiring faithful structural recovery and stable color correction.

4.3.3 Efficiency assessment

Using the computational metrics reported in Table 2, we further analyze the efficiency of competing methods under the same implementation and hardware setting. Overall, FDM-Net achieves a favorable efficiency–performance trade-off. Although it does not attain the lowest MACs or parameter count, FDM-Net remains substantially lighter than several representative baselines, e.g., reducing MACs by 83.6% compared to WaterNet and by 87.8% compared to GUPDM, while cutting parameters by 93.3% relative to CCMSRNet. In terms of practical inference speed, FDM-Net is faster than most learning-based competitors in Table 2. Notably, theoretical complexity does not always translate directly into wall-clock latency. For example, although FIVE A+ and UWCNN have much smaller MACs/Params, their E2E.Time (0.07 s) is still comparable to or slightly higher than that of FDM-Net (0.06 s), suggesting that memory access patterns and operator-level efficiency also affect runtime. These results show that FDM-Net offers competitive cost with fast end-to-end inference, making it suitable for resource-constrained and efficiency-critical underwater vision applications.

4.4 Application test

To demonstrate the effectiveness and applicability of the proposed FDM-Net in practical UIE tasks, we evaluated it across several key tasks, including keypoint detection, edge detection, image segmentation, and salient region detection. For keypoint detection, we used the SIFT algorithm (Lowe 2004). For edge detection, we applied the Canny algorithm (Canny 2009). For image segmentation, the K-means algorithm was used (Lloyd 1982). For salient region detection, Gaussian filtering was employed (Gonzalez 2009). As shown in Fig. 10, the enhanced images exhibited more structured and consistent keypoints. In the edge detection task, the results for the unenhanced images were inaccurate, with many edges undetected. In the image segmentation task, the unenhanced images produced imprecise results, with numerous areas either misclassified or omitted. However, in the enhanced images, the distinction between objects and the background became much clearer. In the salient region detection task, the enhanced images showed an expansion and increase in the areas of interest. FDM-Net not only improved the quality of underwater images but also significantly enhanced the subsequent tasks, including keypoint detection, edge detection, segmentation, and salient region detection. These improvements are crucial for marine robotics and oceanography

Table 2 Computational cost comparison for all methods. The best results are highlighted in red, and the second-best are blue. – indicates that the corresponding MACs/Params are unavailable

Methods	Year	Journal/Conf.	Computational Cost		
			MACs(G) ↓	Params(M) ↓	E2E.Time(s) ↓
WaterNet (Li et al. 2019b)	2019	TIP	71.42	1.09	0.01
UIEC2Net (Wang et al. 2021)	2021	SPIC	26.06	0.53	0.09
SCNet (Liu et al. 2020)	2022	ICASSP	23.56	0.82	0.02
GUPDM (Mu et al. 2023)	2023	ACM MM	95.80	1.49	0.50
Shallow-UWnet (Naik et al. 2021)	2021	AAAI	21.63	0.22	0.07
UWCNN (Li et al. 2020)	2020	PR	2.61	0.04	0.07
CCMSRNet (Qi et al. 2023)	2023	TGRS	43.60	21.13	0.08
USUIR (Fu et al. 2022b)	2022	AAAI	14.81	0.23	0.05
SFGNet (Zhao et al. 2023)	2023	TIP	81.58	1.30	0.02
FIVE A+ (Jiang et al. 2023a)	2023	BMVC	0.58	0.01	0.07
PUIE-Net (Fu et al. 2022a)	2022	ECCV	30.09	1.40	0.11
BMFH-Net (Sun et al. 2025)	2025	TCSVT	16.77	2.42	0.10
GuidedHybSensUIR (Guo et al. 2025)	2025	TCSVT	10.05	1.15	0.11
Ours (FDM-Net)	–	–	11.71	1.41	0.06

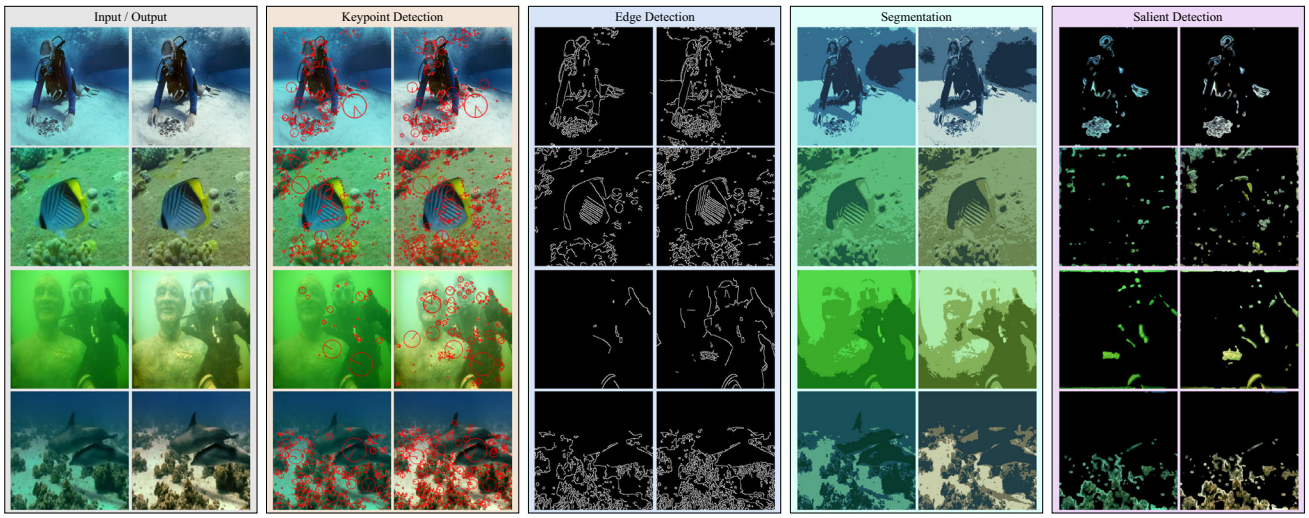


Fig. 10 Application tests in actual underwater environments. From left to right: distorted and enhanced images, keypoint detection results, segmentation outputs, and salient detection results for both distorted and enhanced underwater images

applications, where stable and accurate image analysis plays a vital role in both research and operations.

4.5 Ablation study

To evaluate the contribution of each component in FDM-Net, we perform a systematic ablation study by removing one module or one loss term at a time. The quantitative results on the T-90 dataset are reported in Table 3, while Fig. 11 presents representative visual comparisons together with the corresponding LFD values on a single image. As shown in Table 3 and Fig. 11, the full model achieves the best overall performance, i.e., 23.58 dB PSNR, 0.915 SSIM, and 0.0749 LPIPS, while also obtaining the lowest LFD value of 0.203 on the representative example, which validates the effectiveness of the overall design.

4.5.1 Component terms

We first analyze the contribution of the IDFS. When IDFS is removed, the performance drops to 23.27/0.907/0.0792 in terms of PSNR/SSIM/LPIPS, and the LFD value increases from 0.203 to 0.228. This result indicates that explicitly decomposing degraded features into low- and high-frequency components is beneficial for underwater restoration. Without IDFS, the network loses the frequency-decoupled representation that is essential for separately handling wavelength-dependent attenuation in low-frequency components and scattering-induced corruption in high-frequency details, thereby degrading both global restoration fidelity and local detail recovery.

CB and MSA are used to further encode the low- and high-frequency features, respectively. Removing CB decreases the performance to 23.51/0.910/0.0774, and the corresponding LFD rises to 0.248. Removing MSA yields 23.56/0.910/0.0756 with an LFD of 0.234. Although the drops in PSNR and SSIM are relatively moderate, the noticeable increase in LFD suggests that both modules are important for refining branch-specific representations. In particular, CB helps stabilize low-frequency structural and illumination encoding, while MSA strengthens long-range dependency modeling in the high-frequency branch, which is beneficial for preserving fine textures and improving local detail consistency.

Table 3 Ablation study of individual components and loss functions on the T-90 dataset

Methods	Metrics		
	PSNR ↑	SSIM ↑	LPIPS ↓
-w/o IDFS	23.27	0.907	0.0792
-w/o CB	23.51	0.910	0.0774
-w/o MSA	23.56	0.910	0.0756
-w/o FEGBM	19.42	0.862	0.1150
-w/o HFDE(DERBs)	23.38	0.912	0.0775
-w/o LFDE(MDABs)	23.11	0.911	0.0798
-w/o Gate Network	22.96	0.912	0.0813
-w/o PUIM	23.38	0.912	0.0783
-w/o BAMM	22.95	0.905	0.0793
-w/o MCFN	23.45	0.911	0.0757
-w/o $\mathcal{L}_{spatial}$	23.51	0.912	0.0784
-w/o $\mathcal{L}_{wave}$	23.55	0.910	0.0800
-w/o $\mathcal{L}_{fourier}$	22.84	0.893	0.0902
Full Model	23.58	0.915	0.0749

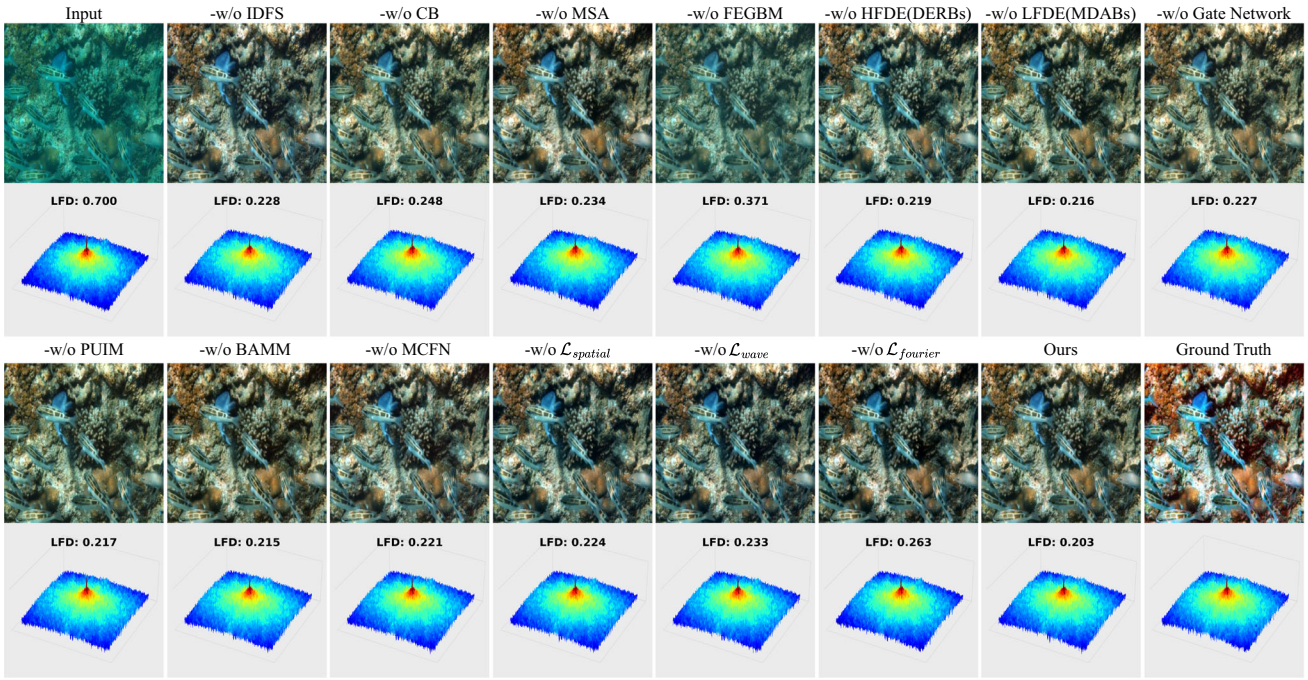


Fig. 11 Visual examples of the ablation experiment on the proposed components

We then evaluate the overall contribution of FEGBM, which is the core restoration module in FDM-Net. Removing FEGBM causes the most severe degradation among all component ablations: PSNR drops from 23.58 to 19.42, SSIM decreases from 0.915 to 0.862, LPIPS increases from 0.0749 to 0.1150, and LFD rises sharply from 0.203 to 0.371. This substantial performance gap demonstrates that FEGBM plays a dominant role in the proposed framework. Without frequency-expert guidance, physics-based inversion, and bidirectional modulation, the network cannot effectively model frequency-specific underwater degradation, resulting in poor structure reconstruction, severe detail loss, and obvious perceptual distortion. To further understand FEGBM, we analyze its internal components. Removing the HFDE(implemented by DERBs) reduces the performance to 23.38/0.912/0.0775, with an LFD of 0.219. This confirms that HFDE is important for recovering scattering-corrupted edges and textures. Removing the LFDE(implemented by MDABs) leads to 23.11/0.911/0.0798 and an LFD of 0.216, indicating that LFDE contributes to preserving global structure, contrast, and illumination consistency. When the Gate Network is removed, the model further drops to 22.96/0.912/0.0813, and the LFD increases to 0.227. This shows that adaptive expert selection is necessary for balancing different expert responses and enhancing the discriminative restoration ability of the frequency-guided branches. For the PUIM inside FEGBM, removing PUIM results in 23.38/0.912/0.0783 with an LFD of 0.217. This suggests that the physics-based inversion mechanism provides physically meaningful correction for underwater attenuation and scattering, leading to more realistic enhancement. Removing BAMM causes a larger performance decline, yielding 22.95/0.905/0.0793 and an LFD of 0.215. This result shows that bidirectional affine modulation is important for cross-frequency interaction after physical inversion. Without BAMM, the network becomes less effective at coordinating global illumination and local textures, which weakens the final restoration quality.

Finally, we evaluate the contribution of MCFN in the upsampling stage. Removing MCFN gives 23.45/0.911/0.0757, and the LFD increases to 0.221. Although the changes in PSNR and SSIM are not as large as those caused by removing FEGBM, the increase in LFD and the visual results indicate that MCFN is important for refining boundaries and textures during upsampling. This observation is consistent with the design of MCFN, which aggregates multi-scale spatial cues to suppress interpolation artifacts and recover fine-grained details.

4.5.2 Loss function terms

We further analyze the contribution of each loss term. Removing the spatial loss $\mathcal{L}_{\text{spatial}}$ results in 23.51/0.912/0.0784, with the LFD increasing to 0.224. This indicates that spatial supervision provides stable pixel-level guidance for color and structure reconstruction, although its contribution is relatively less significant than the frequency-domain constraints. Removing the wavelet loss $\mathcal{L}_{\text{wave}}$ yields 23.55/0.910/0.0800, and the LFD rises to 0.233. This suggests that the wavelet-

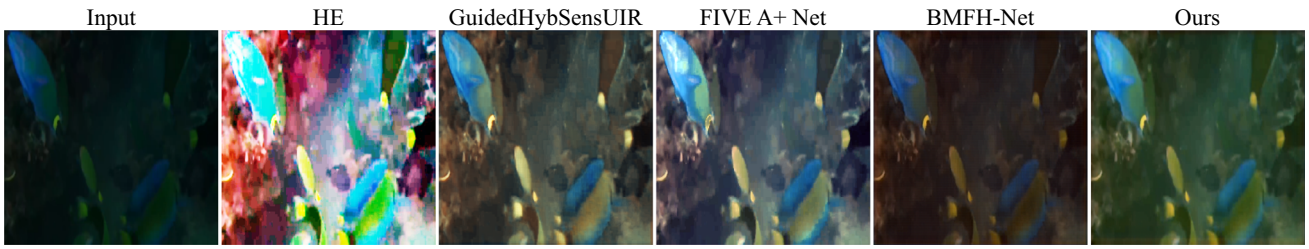


Fig. 12 Failure case of FDM-Net on non-reference Challenge-60 datasets

domain supervision is particularly useful for maintaining low-/high-frequency consistency and multi-scale detail alignment, even though the drop in PSNR is relatively small. In contrast, removing the Fourier loss $\mathcal{L}_{\text{fourier}}$ causes the largest degradation among all loss ablations, reducing the performance to 22.84/0.893/0.0902 and increasing the LFD to 0.263. This clearly demonstrates that Fourier-domain supervision is crucial for constraining spectral consistency, suppressing frequency distortion, and preserving high-frequency structures.

5 Discussion

Although FDM-Net achieves strong overall performance on multiple benchmarks, it still has limitations under extremely challenging underwater conditions. As shown in Fig. 12, when the scene suffers from extremely high turbidity, severe color distortion, and very weak target visibility, the restored results remain unsatisfactory, even though FDM-Net can still improve visual clarity to some extent. In such cases, the recoverable information in the degraded image is already very limited, which fundamentally constrains the final restoration quality.

This performance degradation is closely related to the frequency-decoupled design of FDM-Net. The model assumes that underwater degradation can be separated into low-frequency distortions, such as illumination shift and wavelength-dependent attenuation, and high-frequency corruption, such as scattering-induced texture degradation. This assumption is effective in most common underwater scenes and is supported by the ablation results, where removing IDFS consistently degrades PSNR, SSIM, LPIPS, and LFD. However, under extreme degradation, this assumption becomes less reliable. When high-frequency details are severely suppressed, the high-frequency branch receives weak and noisy inputs, reducing the effectiveness of MSA. Similarly, when low-frequency components are dominated by strong color bias or non-uniform illumination, CB may provide less stable structural guidance. FEGBM remains the most critical module, as its removal causes the largest performance drop. This also indicates that the overall framework strongly depends on the validity of the frequency-guided restoration process implemented within FEGBM. Specifically, FEGBM consists of frequency-expert guidance, the physics-based inversion module PUIM, and the bidirectional interaction module BAMM. When degradation is too severe or deviates significantly from the modeled patterns, these components may not fully capture the true corruption process, thereby limiting the final enhancement quality. In addition, although MCFN helps refine boundaries and textures during upsampling, its benefit is naturally limited when earlier stages fail to preserve sufficient recoverable details.

More broadly, FDM-Net also has several structural limitations as a frequency-based underwater restoration model. First, it does not explicitly model scene depth, water type, or turbidity level, although these factors strongly affect real underwater degradation. Second, the physical inversion in PUIM is still a simplified approximation of underwater image formation and may therefore be insufficient for highly heterogeneous water conditions or unseen domains. Third, although Fourier- and wavelet-domain supervision effectively constrain spectral consistency, they mainly operate at the signal level and do not explicitly incorporate semantic priors. As a result, the model may still struggle in cases involving heavy occlusion, extremely small targets, or severely degraded semantic structures. Overall, FDM-Net is effective when frequency-specific degradation patterns remain observable and recoverable, but its advantage may diminish when degradation becomes excessively severe, highly coupled, or strongly out-of-distribution. In future work, we will explore more robust underwater degradation modeling by introducing explicit turbidity estimation, depth-aware or water-type-adaptive priors, and stronger semantic-guided restoration mechanisms.

6 Conclusion

In this paper, we present FDM-Net, an underwater image enhancement model that integrates frequency decomposition with an underwater imaging physics model, aiming to achieve perceptually more plausible image quality at

both global and local detail levels while maintaining physical interpretability. By leveraging frequency decomposition, FDM-Net effectively models and represents texture and edge features in underwater images, thereby narrowing the gap between degraded observations and their corresponding clear counterparts. In addition, the physics-guided feature inversion and bidirectional affine modulation establish a transparent link between the network representations and the underlying imaging process, further enhancing the interpretability of the enhancement results. However, the proposed method still exhibits certain limitations when enhancing extremely challenging underwater scenes. In future work, we will focus on further improving the performance, robustness, and interpretability of FDM-Net under such extreme conditions.

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Author Contributions Yu.Ma was responsible for all aspects of the research, including designing the study, conducting the experiments, analyzing the data, and writing the manuscript Qiuling Yang and Rongxin Zhu provided guidance throughout the study, reviewed the manuscript, and offered valuable suggestions for improvement.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval This study does not involve human participants or animals. Ethical approval was not required.

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